PVsyst V8.1.3

PVsyst - Simulation report

Grid-Connected System

Project: BATTERY (3)

Variant: 12.6

No 3D scene defined, no shadings

System power: 12.60 kWp

Kampung Chuang - Malaysia

Author

NORTHERN SOLAR SDN. BHD. (Malaysia)

Project summary

Geographical Site	Situation		Project settings	
Kampung Chuang	Latitude	3.4869 °(N)	Albedo	0.20
Malaysia	Longitude	101.6284 °(E)		
	Altitude	22 m		
	Time zone	UTC+8		

Weather data
Kampung Chuang
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System
Simulation for year no 1

No 3D scene defined, no shadings

Orientation #1		Orientation #2		Orientation #3	
Fixed plane		**Fixed plane**		**Fixed plane**	
Tilt/Azimuth	20 / 90 °	Tilt/Azimuth	20 / -90 °	Tilt/Azimuth	20 / 0 °

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	20 units	Nb. of units	1 unit
Pnom total	12.60 kWp	Total power	10 kWac
		Pnom ratio	1.26

Results summary

Simulation step	Hourly				
Produced Energy	17255 kWh/year	Specific production	1369 kWh/kWp/year	Perf. Ratio PR	79.37 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step
Hourly

Orientation #1		Orientation #2		Orientation #3	
Fixed plane		**Fixed plane**		**Fixed plane**	
Tilt/Azimuth	20 / 90 °	Tilt/Azimuth	20 / -90 °	Tilt/Azimuth	20 / 0 °

Models used		Horizon		Near Shadings
Transposition	Perez	Average Height	2.6 °	no Shadings
Diffuse	Perez, Meteonorm			
Circumsolar	separate			

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module		Inverter	
Manufacturer	Astronergy	Manufacturer	Hoymiles
Model	CHSM66RN(DG)/F-BH-630	Model	HIT-10L-G3
(Custom parameters definition)		(Custom parameters definition)	
CHSM66RN(DG)F-BH-630.PAN		Unit Nom. Power	10.00 kWac
Unit Nom. Power	630 Wp		

Array #1 - PV Array

Orientation	1		
Tilt/Azimuth	20/90 °		
Number of PV modules	10 units	Number of inverters	1 * MPPT 33% 0.3 unit
Nominal (STC)	6.30 kWp	Total power	3.3 kWac
Modules	1 unit x 10 In series		
At operating cond. (50°C)		Operating voltage	125-900 V
Pmpp	5.88 kWp	Pnom ratio (DC:AC)	1.89
U mpp	377 V		
I mpp	16 A		

Array #2 - Sub-array #2

Orientation	2		
Tilt/Azimuth	20/-90 °		
Number of PV modules	5 units	Number of inverters	1 * MPPT 33% 0.3 unit
Nominal (STC)	3150 Wp	Total power	3.3 kWac
Modules	1 unit x 5 In series		
At operating cond. (50°C)		Operating voltage	125-900 V
Pmpp	2940 Wp	Pnom ratio (DC:AC)	0.95
U mpp	189 V		
I mpp	16 A		

Array #3 - Sub-array #3

Orientation	3		
Tilt/Azimuth	20/0 °		
Number of PV modules	5 units	Number of inverters	1 * MPPT 33% 0.3 unit
Nominal (STC)	3150 Wp	Total power	3.3 kWac
Modules	1 unit x 5 In series		
At operating cond. (50°C)		Operating voltage	125-900 V
Pmpp	2940 Wp	Pnom ratio (DC:AC)	0.95
U mpp	189 V		
I mpp	16 A		

PV Array Characteristics

Total PV power		**Total inverter power**	
Nominal (STC)	13 kWp	Total power	10 kWac
Total	20 modules	Number of inverters	1 unit
Module area	54.0 m²	Pnom ratio	1.26
Cell area	50.5 m²	No power sharing	

Array losses

Array Soiling Losses
Loss Fraction 1.5 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

LID - Light Induced Degradation
Loss Fraction 1.45 %

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 1.00 % at MPP

Module average degradation
Year no 1
Loss factor 0.55 %/year
Imp / Vmp contributions 80% / 20%

Mismatch due to degradation
Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

IAM loss factor
ASHRAE Param.: IAM = 1 - bo (1/cosi -1)
bo Param. 0.05

DC wiring losses

Global wiring resistance 10 mΩ
Loss Fraction 2.00 % at STC

Array #1 - PV Array		**Array #2 - Sub-array #2**	
Global array res.	786 mΩ	Global array res.	196 mΩ
Loss Fraction	3.00 % at STC	Loss Fraction	1.50 % at STC
Array #3 - Sub-array #3			
Global array res.	196 mΩ		
Loss Fraction	1.50 % at STC		

System losses

Unavailability of the system
Time fraction 1.0 %
3.7 days,
3 periods

AC wiring losses

Inv. output line up to injection point
Inverter voltage 400 Vac tri
Loss Fraction 1.00 % at STC

Global System
Wire section Copper 3 x 4 mm²
Wires length 28 m

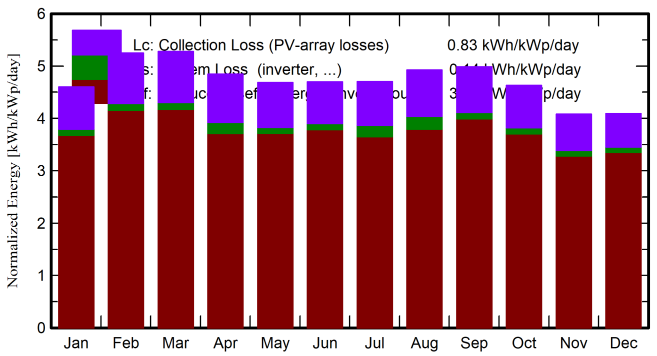
Main results

Simulation step
Hourly

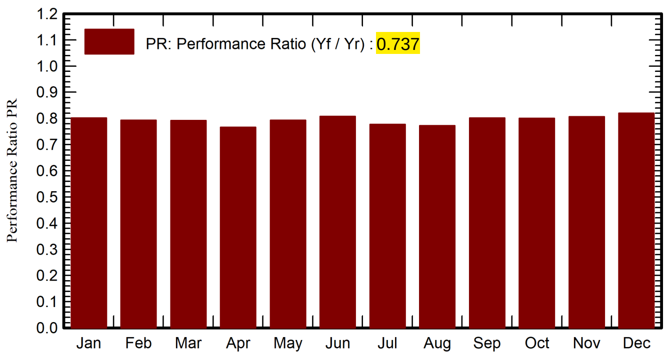
System Production

Produced Energy	17255 kWh/year	Specific production	1369 kWh/kWp/year
		Perf. Ratio PR	79.37 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	141.8	70.84	26.85	142.6	135.9	1484	1439	0.801
February	149.0	72.94	27.46	146.9	140.4	1513	1468	0.793
March	167.7	87.92	27.68	163.5	156.3	1682	1632	0.792
April	151.7	74.54	27.73	145.3	138.5	1484	1404	0.767
May	153.4	76.58	28.05	145.3	138.2	1497	1452	0.793
June	150.5	73.96	27.96	140.9	133.8	1476	1433	0.807
July	155.4	74.21	27.56	145.7	138.7	1514	1427	0.777
August	160.8	80.04	27.59	152.6	145.5	1580	1485	0.773
September	154.9	82.48	27.18	149.6	142.7	1556	1511	0.801
October	146.2	81.34	27.07	143.6	136.7	1494	1449	0.801
November	122.9	70.64	26.68	122.4	116.3	1282	1243	0.806
December	127.2	79.83	26.64	126.9	120.6	1351	1311	0.820
Year	1781.5	925.32	27.37	1725.4	1643.5	17912	17255	0.794

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

PVsyst V8.1.3
VC0, Simulation date:
with V8.1.3

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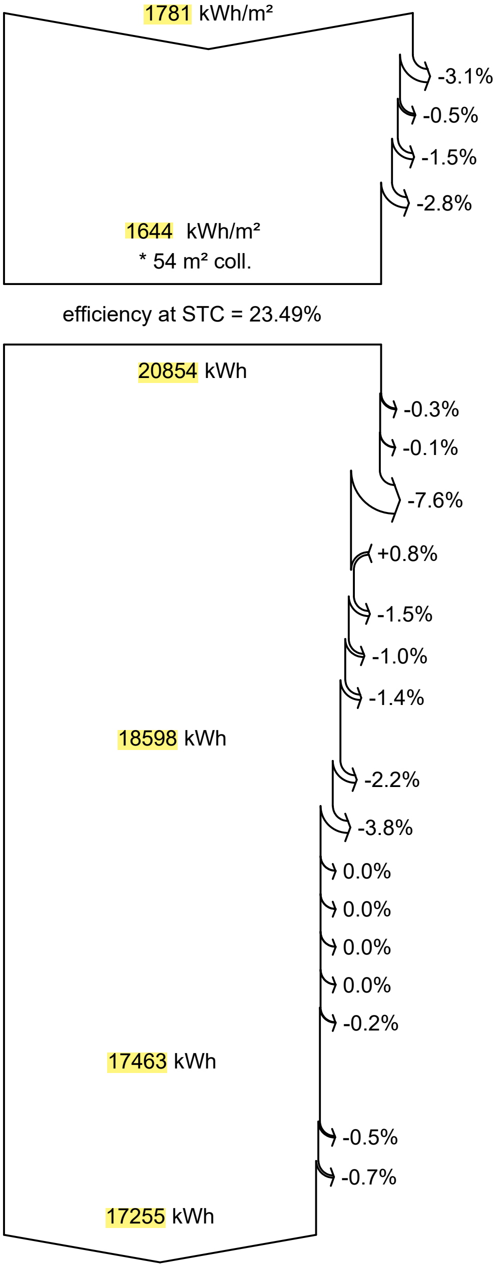
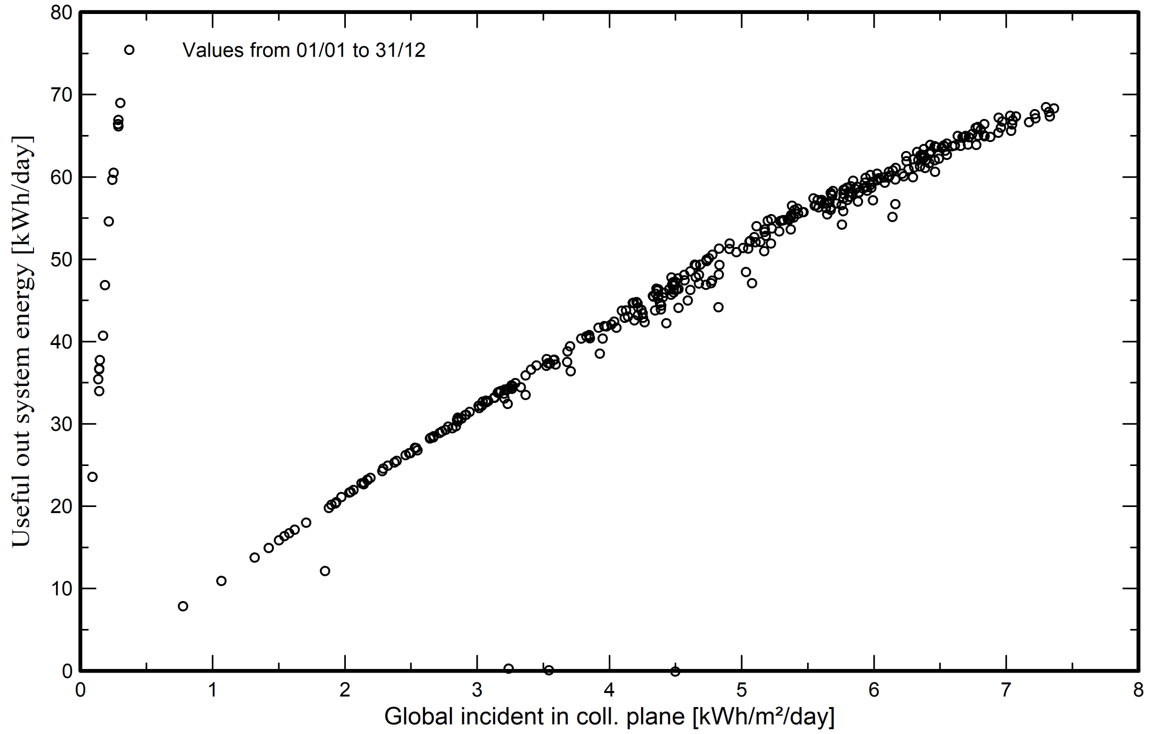
Loss diagram



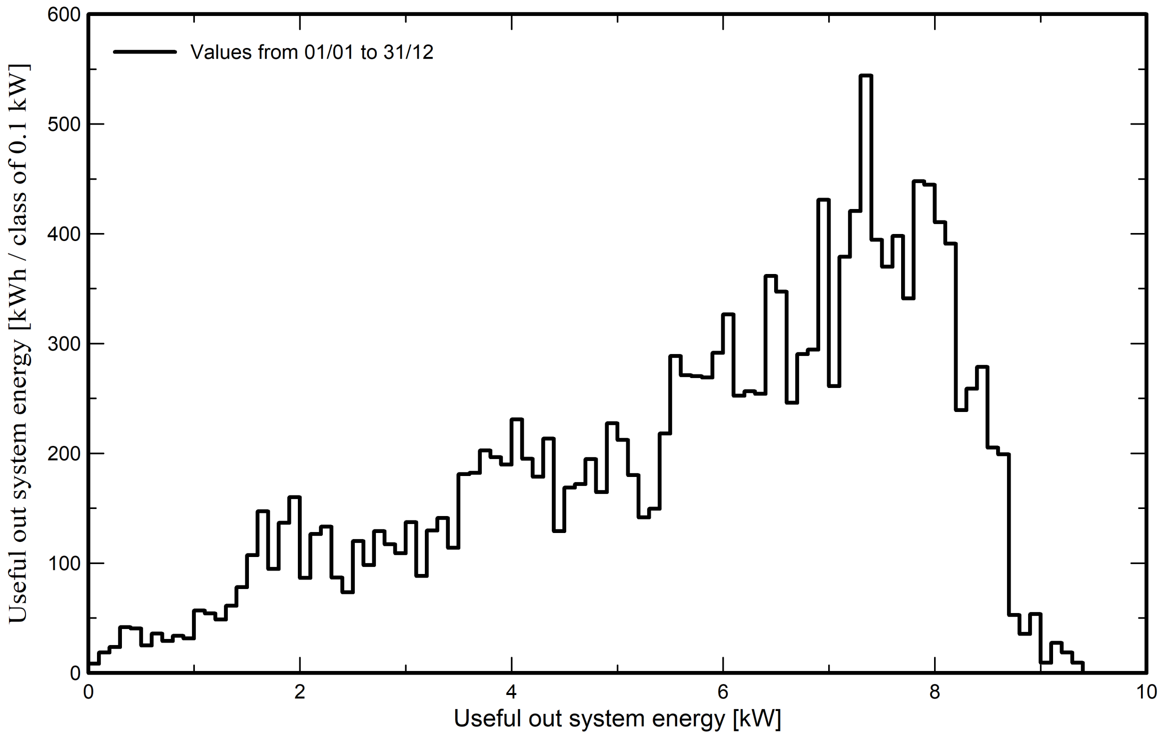
1781 kWh/m²	**Global horizontal irradiation**
-3.1%	**Global incident in coll. plane**
-0.5%	Far Shadings / Horizon
-1.5%	Soiling loss factor
-2.8%	IAM factor on global
1644 kWh/m² * 54 m² coll.	**Effective irradiation on collectors**
efficiency at STC = 23.49%	PV conversion
20854 kWh	**Array nominal energy (at STC effic.)**
-0.3%	Module Degradation Loss (for year #1)
-0.1%	PV loss due to irradiance level
-7.6%	PV loss due to temperature
+0.8%	Module quality loss
-1.5%	LID - Light induced degradation
-1.0%	Module array mismatch loss
-1.4%	Ohmic wiring loss
18598 kWh	**Array virtual energy at MPP**
-2.2%	Inverter Loss during operation (efficiency)
-3.8%	Inverter Loss over nominal inv. power
0.0%	Inverter Loss due to max. input current
0.0%	Inverter Loss over nominal inv. voltage
0.0%	Inverter Loss due to power threshold
0.0%	Inverter Loss due to voltage threshold
-0.2%	Night consumption
17463 kWh	**Available Energy at Inverter Output**
-0.5%	AC ohmic loss
-0.7%	System unavailability
17255 kWh	**Energy injected into grid**

Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



P50 - P90 evaluation

Weather data

Source	Meteonorm 9.0 dll, Sat=100%
Kind	Specific year
Year	Synthetic
Year-to-year variability(Variance)	4.0 %
Specified Deviation	
Year deviation from average	0.0 %

Global variability (weather data + system)

Variability (Quadratic sum)	5.9 %

Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %
Custom variability	4.0 %

Annual production probability

Variability	1024 kWh
P50	17255 kWh
P90	15942 kWh
P75	16565 kWh

Probability distribution